

Seon-Ho Lee

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EDUCATION

Korea University

Ph.D. in Electrical Engineering (Adv. Chang-Su Kim)

M.S. in Electrical Engineering (Adv. Chang-Su Kim)

B.S. in Electrical Engineering (incl. mandatory military service, 2014 - 2016)

Seoul, Korea

Mar. 2020 - Aug. 2024

Mar. 2018 - Feb. 2020

Mar. 2011 - Feb. 2018

RESEARCH INTEREST

Multimodal Foundation Models, Vision-Language Models, Video Understanding, Efficient Long-Context Modeling, Embedding Space Optimization, Rank Estimation

WORK EXPERIENCE

Amazon AGI

Jun. 2025 - Present

Applied Scientist II

Seattle, WA

- Pre-training of **full-duplex, real-time multimodal LLMs** that listen and speak concurrently, for low-latency streaming speech-to-speech interaction
- Core training recipe — data mix, model architecture, and optimization — and operation of production-scale training runs, including monitoring, debugging, and recovery from training instabilities
- Contributor to **Amazon Nova 2**, a family of multimodal foundation models (Lite, Pro, Omni, Sonic) that reason over text, image, video, and speech with up to 1M-token context (technical report)

Amazon Prime Video

Sep. 2023 - Mar. 2024

Applied Scientist Intern

Seattle, WA

- Fully automated audio description (AD) system for long movies (Published in WACV 2025)
 - * Developed the first fine-grained AD event location detection algorithm
 - * Achieved the best AD generation scores on the various datasets
- Learnable video token merging for efficient long-form video understanding (Published in NeurIPS 2024)
 - * Improved the inference speed of a transformer-based network by 7.5 times with 6.6 times less memory
 - * Achieved the best long-video understanding performances on the various datasets

Amazon Prime Video

Jun. 2022 - Sep. 2022

Applied Scientist Intern

Los Angeles, CA

- Unsupervised and semi-supervised active speaker detection
 - * Reduced the annotation burden for active speaker detection by 50% via curricular learning
 - * Developed the first unsupervised active speaker detection method

Industry Research Projects

Researcher at media communication lab (Korea University)

Seoul, Korea

- [Naver] AI system for automated aesthetic photo editing *Oct. 2021 - Sep. 2022*
- [National Research Foundation of Korea] Static and dynamic point cloud compression *Aug. 2017 - Jan. 2021*
- [Naver Labs] Future motion estimation for autonomous driving *Mar. 2018 - Aug. 2018*

TOP CONFERENCE PUBLICATIONS

- [ICML 2026] Chaewon Lee, **Seon-Ho Lee**, Chang-Su Kim, “Stochastic Order Learning: An Approach to Rank Estimation Using Noisy Data”
- [NeurIPS 2024] **Seon-Ho Lee**, Jue Wang, Zhikang Zhang, David Fan, Xinyu Li, “Video Token Merging for Long Video Understanding”
- [ICLR 2024] **Seon-Ho Lee**, Nyeong-Ho Shin, Chang-Su Kim, “Unsupervised Order Learning”
- [CVPR 2024] Chaewon Lee, **Seon-Ho Lee**, Chang-Su Kim, “MFP: Making Full use of Probability Maps for Interactive Image Segmentation”

- [CVPR 2024] Nyeong-Ho Shin, **Seon-Ho Lee**, Chang-Su Kim, “Blind Image Quality Assessment Based on Geometric Order Learning”
- [NeurIPS 2022] **Seon-Ho Lee**, Nyeong-Ho Shin, Chang-Su Kim, “Geometric Order Learning for Rank Estimation”
- [ECCV 2022] **Seon-Ho Lee**, Chang-Su Kim, “Order Learning Using Partially Ordered Data via Chainization”
- [CVPR 2022 Oral] Nyeong-Ho Shin, **Seon-Ho Lee**, Chang-Su Kim, “Moving Window Regression: A Novel Approach to Ordinal Regression”
- [ICLR 2021] **Seon-Ho Lee**, Chang-Su Kim, “Deep Repulsive Clustering of Ordered Data Based on Order-Identity Decomposition”

OTHER PUBLICATIONS

- [Tech Report 2025] Amazon AGI, “Amazon Nova 2: Multimodal Reasoning and Generation Models”
- [WACV 2025] **Seon-Ho Lee**, Jue Wang, David Fan, Zhikang Zhang, Linda Liu, Xiang Hao, Vimal Bhat, Xinyu Li, “Now You See Me: Context-Aware Automatic Audio Description”
- [JVCIR 2024] Nyeong-Ho Shin, **Seon-Ho Lee**, Chang-Su Kim, “Image Cropping Based on Order Learning”
- [ICASSP 2023 Challenge Winner] Ju-Hyung Lee, Joohan Lee, **Seon-Ho Lee**, Andreas F. Molisch, “PMNet: Large-Scale Channel Prediction System for ICASSP 2023 First Pathloss Radio Map Prediction Challenge”
- [APSIPA TSIP 2023] **Seon-Ho Lee**, Nyeong-Ho Shin, Chang-Su Kim, “Order Learning — An Overview”
- [APSIPA ASC 2022] Jin-won Ko, Nyeong-Ho Shin, **Seon-Ho Lee**, Chang-Su Kim, “Unified Angle Adjustment Network for Image Composition Enhancement”
- [JVCIR 2021] **Seon-Ho Lee**, Chang-Su Kim, “SAF-Nets: Shape-Adaptive Filter Networks for 3D Point Cloud Processing”
- [Access 2019] **Seon-Ho Lee**, Han-Ul Kim, Chang-Su Kim, “ELF-Nets: Deep Learning on Point Clouds Using Extended Laplacian Filter”

HONORS & AWARDS

Brain Korea 21 Fellowship, National Research Foundation of Korea	<i>Mar. 2018 - Fall 2023</i>
Best Academic Award, Korea University	<i>Spring 2017, Fall 2017</i>
Full Tuition Scholarship, Korea Student Aid Foundation	<i>Mar. 2011</i>

PROFESSIONAL SERVICES

Conference Reviewer: CVPR (2022–2026), ICCV (2023, 2025), ECCV (2022, 2024), NeurIPS (2023–2025), ICLR (2024–2026), ICML 2025, AAAI (2025, 2026), WACV (2025, 2026), AISTATS 2025

Journal Reviewer: IEEE TIP, IEEE TMM, IEEE TCSVT, JVCIR, AIR, VRIH

Standards: Member of ISO/IEC JTC1/SC29/WG11 (MPEG)

Lecture: Seoul School of Integrated Sciences & Technologies *May and Nov. 2020*

SKILLS

Python, PyTorch, JAX, TensorFlow, MATLAB